

Large Language Models & Generative AI

LLMs, Prompt Engineering, and Capabilities

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Algorithm Basics and BW

What We'll Cover Today

1. 🧠 What are LLMs?
2. 🤖 How LLMs work (simplified)
3. ✎ Generative AI capabilities
4. ✎ Prompt engineering
5. ⚠️ Limitations & risks
6. 💻 Hands-on: Prompt workshop

Quick Recap: AI Fundamentals

Last class we learned:

- ▶ 💡 AI definition & types
- ▶ ⚙️ Machine learning basics
- ▶ 📈 AI vs ML vs Deep Learning
- ▶ 💼 Business use cases
- ▶ 🤖 First interaction with AI
- ▶ ✓ AI capabilities & limits

Today: Deep dive into LLMs - the AI everyone is talking about

What Are Large Language Models (LLMs)?

Definition:

- ▶  AI trained on massive text data
- ▶  "Read" billions of web pages, books
- ▶  Understand & generate human language
- ▶  Answer questions, write content
- ▶  Predict next word in sequence
- ▶  Neural networks (billions of parameters)
- ▶  Seem intelligent (pattern matching)
- ▶  Multi-language support

Key insight: Not truly "understanding" - statistical pattern prediction

Major LLMs in 2024-2025

Consumer LLMs:

- ▶  **ChatGPT** (OpenAI)
 - ▶ GPT-4, GPT-4o
 - ▶ 200M+ users
- ▶  **Gemini** (Google)
 - ▶ Multimodal
 - ▶ Deep search integration
- ▶  **Claude** (Anthropic)
 - ▶ Long context
 - ▶ Safety-focused
 - ▶ Ethical & Constitutional

Open Source & Others:

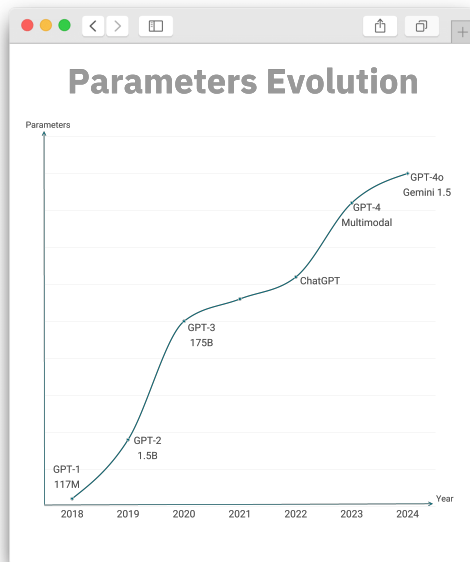
- ▶  **Llama** (Meta)
 - ▶ Free, customizable
 - ▶ Run locally
- ▶  **Copilot** (Microsoft)
 - ▶ Office integration
 - ▶ Enterprise focus
- ▶  **Specialized:**
 - ▶ Code: GitHub Copilot
 - ▶ Legal: Harvey AI
 - ▶ Medical: Med-PaLM

LLM Evolution Timeline

Key milestones:

- ▶ 🗓️ **2018:** GPT-1 (117M params)
- ▶ ⬆️ **2019:** GPT-2 (1.5B params)
- ▶ 🚀 **2020:** GPT-3 (175B params)
- ▶ 🔥 **2022:** ChatGPT launch
- ▶ ★ **2023:** GPT-4 (multimodal)
- ▶ ⚡ **2024:** GPT-4o, Gemini 1.5

Trend: Bigger, faster, smarter



How LLMs Work - Simplified (No Math!)

5-step process:

1.  **Input:** You type a prompt
 - ▶ Text broken into "tokens" (words/parts)
 - ▶ Converted to numbers
2.  **Processing:** Neural network analyzes
 - ▶ Billions of parameters (learned patterns)
 - ▶ Finds relevant patterns from training
3.  **Prediction:** Calculates next word probabilities
 - ▶ "The sky is ___" → "blue" (high probability)
 - ▶ Multiple possibilities considered

How LLMs Work - Simplified (Continued)

4. **Sampling:** Chooses word (with some randomness)

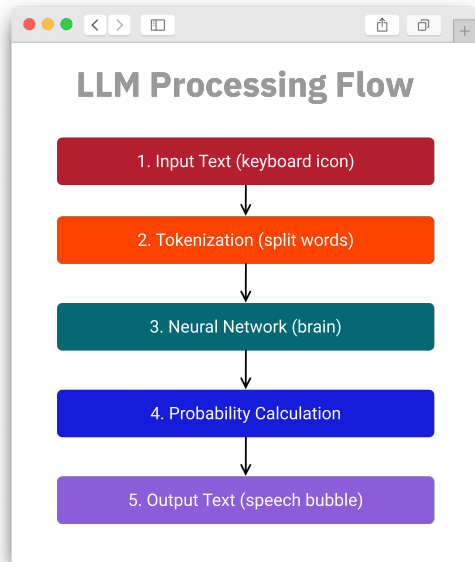
- ▶ "Temperature" controls creativity
- ▶ Higher temp = more random/creative
- ▶ Lower temp = more predictable

5. **Repeat:** Generates word by word

- ▶ Each word influences next
- ▶ Continues until complete thought
- ▶ Can generate thousands of words

Key takeaway: LLMs don't "think" - they predict patterns!

LLM Architecture - Simplified



Three key components:

- ▶  **Transformer:** Core architecture
 - ▶ Attention mechanism
 - ▶ Context understanding
- ▶  **Parameters:** Learned patterns
 - ▶ Billions of numbers
 - ▶ From training data
- ▶  **Context window:** Memory
 - ▶ How much it "remembers"
 - ▶ 4K to 1M+ tokens

Training vs Fine-Tuning vs Inference

Training:

- ▶  Learn from massive datasets
- ▶  Months, millions of dollars
- ▶  Done by big companies
- ▶  You don't do this

Fine-Tuning:

- ▶  Adapt pre-trained model
- ▶  Hours to days
- ▶  Affordable (\$100s)
- ▶  Companies can do this

Inference:

- ▶  Using the model
- ▶  Seconds per response
- ▶  Cheap (pennies)
- ▶  What you do daily

Analogy:

- ▶ Training = Building factory
- ▶ Fine-tuning = Customizing
- ▶ Inference = Using product

What is Generative AI? AI that creates new content

Text Generation:

- ▶  Articles, reports, emails
- ▶  Code, scripts
- ▶  Translations
- ▶  Conversations

Image Generation:

- ▶  DALL-E, Midjourney, Stable Diffusion
- ▶  Logos, illustrations, photos
- ▶  Image editing & variations

Video & Audio:

- ▶  Sora, Runway (video)
- ▶  AI music generation
- ▶  Voice cloning
- ▶  Text-to-speech

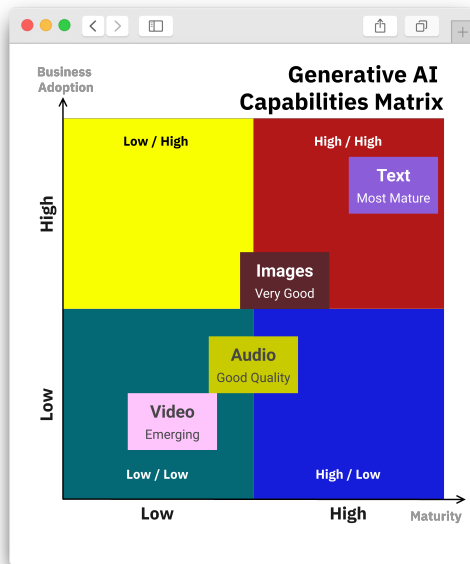
Other:

- ▶  3D models
- ▶  Game assets
- ▶  Molecule design (pharma)

Generative AI Capabilities Matrix

By content type:

- ▶ **Text:** Most mature
 - ▶ High quality
 - ▶ Business-ready
- ▶ **Images:** Very good
 - ▶ Creative, realistic
 - ▶ Still some artifacts
- ▶ **Video:** Emerging
 - ▶ Impressive demos
 - ▶ Not consistent yet
- ▶ **Audio:** Good quality
 - ▶ Music, voices
 - ▶ Copyright concerns



LLMs vs Traditional AI

Aspect	Traditional AI	LLMs/Gen AI
Task scope	Narrow, specific task	Broad, multi-purpose
Training	Labeled data needed	Learns from raw text
Flexibility	Rigid, rule-based	Adaptable, creative
Output	Predictions, labels	New content created
User interface	Technical integration	Natural language
Examples	Spam filter, recommender	ChatGPT, image gen

Key difference: LLMs are generalists, not specialists

What is Prompt Engineering? Crafting inputs to get better AI outputs

Why it matters:

- ▶  Better results
- ▶  Save time, fewer retries
- ▶  More accurate responses
- ▶  Lower API costs
- ▶  Valuable skill for business

What makes a good prompt:

- ▶  Clear & specific
- ▶  Context provided
- ▶  Role defined
- ▶  Format specified
- ▶  Constraints stated

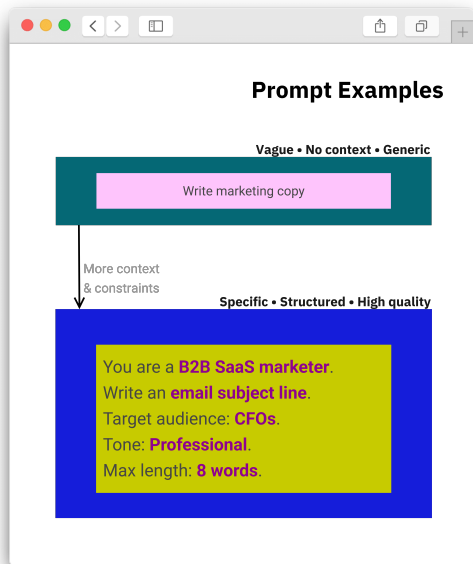
Think of AI as intern: Give clear instructions, context, examples

Prompt Engineering Framework

CRAFT Framework:

1.  **Context:** Set the scene
 - ▶ “You are a financial analyst...”
2.  **Role:** Define AI's role
 - ▶ “...expert in quarterly reports...”
3.  **Action:** What you want done
 - ▶ “...analyze this data and identify trends...”
4.  **Format:** How output should look
 - ▶ “...present as bullet points with 3 key findings...”
5.  **Tone/constraints:** Style & limits
 - ▶ “...keep professional tone, max 200 words”

Prompt Examples: Bad vs Good



Bad prompt ✖:

- ▶ "Write report"
- ▶ Vague
- ▶ No context
- ▶ No format
- ▶ Generic output

Good prompt ✔:

- ▶ Specific task defined
- ▶ Context provided
- ▶ Format specified
- ▶ Constraints clear
- ▶ Quality output

Advanced Prompting Techniques

Chain-of-Thought:

- ▶  Ask AI to “think step-by-step”
- ▶  Shows reasoning
- ▶  Better accuracy
- ▶  Example:
“Let’s solve this step by step:”

Few-Shot Learning:

- ▶  Provide examples
- ▶  AI learns pattern
- ▶  Mimics style
- ▶  “Input→Output” pairs

Role Prompting:

- ▶  “You are a {role}...”
- ▶  Expert consultant
- ▶  Teacher, coach
- ▶  Devil’s advocate

Iterative Refinement:

- ▶  Start broad, refine
- ▶  Ask for improvements
- ▶  “Make it more...”
- ▶  Iterate to perfection

Prompt Templates for Business

Ready-to-use templates:

1. **Email writing:**

- ▶ "Draft professional email to {recipient} about {topic}. Tone: {formal/casual}. Length: {X} sentences."

2. **Data analysis:**

- ▶ "Analyze this data: {paste data}. Identify top 3 trends. Format: Bullet points with evidence."

3. **Summarization:**

- ▶ "Summarize this {document/meeting notes} in {X} sentences. Focus on action items."

Prompt Templates (Continued)

4. **Content creation:**

- ▶ "Create {content type} for {audience} about {topic}. Include: {key points}. Tone: {style}."

5. **Decision support:**

- ▶ "List pros and cons of {decision}. Consider: {factors}. Format: Table."

6. **Brainstorming:**

- ▶ "Generate {X} ideas for {problem}. Target: {audience}. Constraints: {budget/time}."

Pro tip: Save your best prompts as templates!

LLM Limitations You Must Know

Hallucinations:

- ▶  Makes up false info
- ▶  Fake citations, stats
- ▶  Always verify facts
- ▶  Doesn't know when unsure

Knowledge cutoff:

- ▶  Training data has date limit
- ▶  No recent events
- ▶  Varies by model
- ▶  GPT-4: Apr'23, Claude: Aug'23

Biases:

- ▶  Reflects training data biases
- ▶  Cultural, political skews
- ▶  Gender stereotypes
- ▶  Be aware, cross-check

Context limits:

- ▶  Finite memory window
- ▶  Can't process huge docs
- ▶  Loses thread in long chats
- ▶  May need to restart

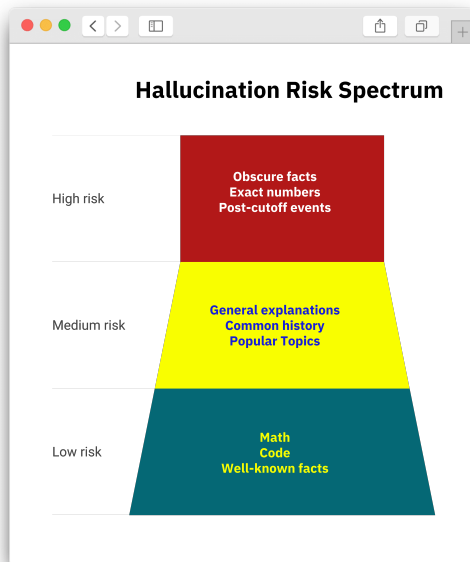
Understanding AI Hallucinations

What causes them:

- ▶ 🎲 Pattern completion priority
- ▶ ❓ No fact-checking mechanism
- ▶ 🧩 Fills gaps with plausible text
- ▶ 📊 Statistical prediction, not truth

How to reduce:

- ▶ ✅ Ask for sources
- ▶ ⚠️ "If unsure, say so"
- ▶ 🗄️ Provide context/data
- ▶ 🔍 Verify externally



Privacy & Security Concerns

What gets stored:

- ▶  Your prompts
- ▶  AI responses
- ▶  Account metadata
- ▶  Potentially indefinitely

What companies do:

- ▶  May retrain on data
- ▶  Improve models
- ▶  Some offer opt-out
- ▶  Enterprise = more control

What NOT to share:

- ▶  Passwords, API keys
- ▶  Financial data
- ▶  Personal identifiable info
- ▶  Proprietary business data
- ▶  Confidential documents

Best practices:

- ▶  Anonymize data
- ▶  Use private/local models
- ▶  Check terms of service

Ethical Considerations

Transparency:

- ▶  Disclose AI use
- ▶  Don't claim as human work
- ▶  Review & edit outputs
- ▶  Take responsibility

Copyright & plagiarism:

- ▶  AI outputs: Murky legal status
- ▶  Training data:
Copyright debates
- ▶  Modify AI outputs significantly
- ▶  Make it your own

Bias & fairness:

- ▶  Audit for bias
- ▶  Consider impact on groups
- ▶  Don't use for sensitive decisions alone
- ▶  Human oversight required

Environmental:

- ▶  Training = massive energy
- ▶  Each query = carbon cost
- ▶  Use efficiently
- ▶  Weigh benefits vs cost

Common Mistakes to Avoid

1.  **Blind trust:** Always verify outputs
2.  **Vague prompts:** Be specific, provide context
3.  **No iteration:** Refine prompts, don't give up
4.  **Copy-paste without review:** Edit and personalize
5.  **Sharing sensitive data:** Protect privacy
6.  **Expecting perfection:** AI is tool, not oracle
7.  **Over-reliance:** Use for augmentation, not replacement

Golden rule: AI enhances human work, doesn't replace judgment

LLM Best Practices Summary

Do's:

- ▶ ✓ Be specific
- ▶ ✓ Verify facts
- ▶ ✓ Iterate prompts
- ▶ ✓ Edit outputs
- ▶ ✓ Protect data
- ▶ ✓ Disclose use

Don'ts:

- ▶ ✗ Blindly trust
- ▶ ✗ Share secrets
- ▶ ✗ Claim as 100% human
- ▶ ✗ Use for final decisions

Hands-On: Prompt Engineering Workshop

Now let's practice!

What we'll do:

- ▶  Open ChatGPT or Claude
- ▶  Write prompts together
- ▶  Compare outputs
- ▶  Improve iteratively
- ▶  Share best prompts

Goal: Learn by doing - see what works!

3 exercises:

1. Email to client
2. Data analysis summary
3. Business idea brainstorm

Time: 15 minutes total

- ▶ 5 min per exercise

Exercise 1: Professional Email

Scenario: Write email to client about project delay

Your task:

1.  Write prompt for AI
2.  Generate email
3.  Review output
4.  Refine prompt
5.  Compare with peers

Include in prompt:

- ▶  Role (you = project manager)
- ▶  Reason for delay
- ▶  New timeline
- ▶  Apologetic tone
- ▶  Format: ~200 words

Exercise 2: Data Analysis Summary

Scenario: Analyze sales data and present findings

Sample data provided:

- ▶  Q4 sales by region
- ▶  Growth percentages
- ▶  Revenue figures
- ▶  Product categories

Include in prompt:

- ▶  Identify top 3 trends
- ▶  What's growing
- ▶  What's declining
- ▶  Format: Bullet points
- ▶  Actionable insights

Exercise 3: Business Idea Brainstorm

Scenario: Generate ideas for campus food delivery service

Your task:

- ▶ 💡 10 creative ideas
- ▶ 👥 Target: Students
- ▶ 💰 Budget-conscious
- ▶ ⌚ Fast delivery
- ▶ 🌱 Sustainability focus

Prompt should ask for:

- ▶ ⅓☰ Numbered list
- ▶ 💬 Brief explanation each
- ▶ ⚖️ Mix of realistic & bold
- ▶ ★ Highlight top 3
- ▶ ! Unique differentiators

Workshop Debrief

Let's discuss what you learned:

Questions to consider:

- ▶  What worked well?
- ▶  What didn't work?
- ▶  Any surprises?
- ▶  Aha moments?
- ▶  How did iteration help?

Key takeaways:

- ▶  Specificity matters
- ▶  Context improves output
- ▶  Iteration is normal
- ▶  Different approaches work
- ▶  Practical business value

Key Takeaways

1. 🧠 LLMs predict patterns, don't truly "understand"
2. 🤖 Major players: ChatGPT, Gemini, Claude, Llama
3. ✎ Generative AI creates text, images, video, audio
4. 🖋 Prompt engineering = valuable business skill
5. ⚠ Limitations: Hallucinations, biases, cutoffs
6. 🛡 Privacy & ethics matter - be responsible
7. ✅ Always verify, iterate, and maintain oversight

Remember: AI is powerful tool for augmentation, not replacement

Questions?